



**ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY
GUWAHATI**

**Course Structure and Syllabus
(From Academic Session 2018-19 onwards)**

B.TECH

**ELECTRONICS AND COMMUNICATION ENGINEERING/
ELECTRONICS AND TELECOMMUNICATION ENGINEERING**

8th SEMESTER



ASSAM SCIENCE AND TECHNOLOGY UNIVERSITY

Course Structure

(From Academic Session 2018-19 onwards)

B.Tech 8th Semester: Electronics and Communication/Telecommunication Engineering

Semester VIII/ B.TECH/ECE/ETE

Sl. No	Sub-Code	Subject	Hours per Week			Credit	Marks	
			L	T	P	C	CE	ESE
Theory								
1	ECE181801	Antenna and Wave Propagation	3	0	0	3	30	70
2	ECE181802	Mobile Communication	3	0	0	3	30	70
3	ECE1818PE3*	Program Elective-3	3	0	0	3	30	70
4	ECE1818OE3*	Open Elective-3	3	0	0	3	30	70
Practical								
1	ECE181822	Project-2	0	0	16	8	100	50
2	ECE181823	General Viva-III	-	-	-	1	-	50
TOTAL			12	0	16	21	220	380
Total Contact Hours per week : 28								
Total Credit: 21								

PROGRAM ELECTIVE – 3		
Sl.No.	Subject Code	Subject
1	ECE1818PE31	Mixed Signal Design
2	ECE1818PE32	Low Power VLSI
3	ECE1818PE33	Biomedical Electronics
4	ECE1818PE33	Any other subject offered from time to time with the approval of the University

OPEN ELECTIVE – 3		
Sl.No.	Subject Code	Subject
1	ECE1818OE31	Machine Learning
2	ECE1818OE32	Advanced Computer Architecture
3	ECE1818OE33	Any other subject offered from time to time with the approval of the University

Detail Syllabus:

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE181801	Antenna and Wave Propagation	3-0-0	3

Course Outcomes:

At the end of the course, students will be able to:

CO1: Apply principles of electromagnetic waves to explain antenna radiation.

CO2: Identify basic antenna parameters

CO3: Design and analyze antenna arrays

CO4: Design practical antennas for both HF and Microwave range

CO5: Identify the characteristics of radio-wave propagation

MODULE 1: Review of Electromagnetic Theory [5 hours]

Maxwell's Equations, wave equations, solutions, plane waves, poynting vector, power flow and potentials.

MODULE 2: Fundamental Concepts [8 hours]

Physical concept of radiation, Radiation pattern, near- and far-field regions, reciprocity, directivity and gain, effective aperture, effective length, polarization, input impedance, radiation resistance, efficiency, beamwidth, Friis transmission equation, Radiation from Wires and Loops- Infinitesimal dipole, finite-length dipole, linear elements near conductors, dipoles for mobile communication, small circular loop.

MODULE 3: Antenna Arrays [7 hours]

Types of arrays: endfire array, broadside array, collinear array, parasitic array

Array analysis: Analysis of uniformly spaced arrays with uniform and non-uniform excitation amplitudes, pattern multiplication

Array synthesis: synthesis of antenna arrays using Schelkunoff polynomial method, binomial DolphChebyshev method and Woodward-Lawson method.

MODULE 4: Practical Antennas-I [5 hours]

HF, VHF and UHF antennas: Folded dipole, V- antenna, Inverted V-antenna, Yagi-Uda antenna. Rhombic antenna, Log-periodic, Loop antenna and Helical antenna. Their characteristics, typical applications, radiation pattern and design procedure.

MODULE 5: Practical Antennas-II [5 hours]

Huygens' principle, radiation from rectangular and circular apertures, design considerations, Babinet's principle, Radiation from sectoral and pyramidal horns, design concepts, prime-focus parabolic reflector and cassegrain antennas.

Micro strip Antennas- Basic characteristics of micro strip antennas, feeding methods, methods of analysis, design of rectangular and circular patch antennas.

MODULE 6: Basic Concepts of Smart Antennas [5 hours]

Concept and benefits of smart antennas, fixed weight beam forming basics, Adaptive beam forming.

MODULE 7: Wave Propagation**[5 hours]**

Ground wave, surface wave, and space wave propagation; Structure of ionosphere and ionospheric propagation; MUF, virtual height, critical frequency, skip distance, Fading Diversity schemes.

Text/Reference Books:

1. J.D. Kraus, Antennas, McGraw Hill, 1988.
2. C.A. Balanis, Antenna Theory - Analysis and Design, John Wiley, 1982.
3. R.E. Collin, Antennas and Radio Wave Propagation, McGraw Hill, 1985.
4. R.C. Johnson and H. Jasik, Antenna Engineering Handbook, McGraw Hill, 1984.
5. I.J. Bahl and P. Bhartia, Micro Strip Antennas, Artech House, 1980.
6. R.K. Shevgaonkar, Electromagnetic Waves, Tata McGraw Hill, 2005
7. R.E. Crompton, Adaptive Antennas, John Wiley

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE181802	Mobile Communication	3-0-0	3

Course Outcomes:

At the end of the course, students will be able to:

CO1: Describe cellular concepts in mobile communication.

CO2: Identify various propagation effects.

CO3: Classify multiple access techniques in mobile communication.

CO4: Analyze various methodologies to improve the cellular capacity

CO5: Outline cellular mobile communication standards.

MODULE 1: Cellular concepts

[5 hours]

Cell structure, frequency reuse, cell splitting, channel assignment, handoff, interference, capacity, power control; Wireless Standards: Overview of 2G and 3G cellular standards.

MODULE 2: Signal propagation

[5 hours]

Propagation mechanism- reflection, refraction, diffraction and scattering, large scale signal propagation and lognormal shadowing.

Fading channels-Multipath and small scale fading- Doppler shift, statistical multipath channel models, narrowband and wideband fading models, power delay profile, average and rms delay spread, coherence bandwidth and coherence time, flat and frequency selective fading, slow and fast fading, average fade duration and level crossing rate. Capacity of flat and frequency selective channels.

MODULE 3: Antennas

[4 hours]

Antennas for mobile terminal monopole antennas, PIFA, base station antennas and arrays.

MODULE 4: Multiple access schemes

[4 hours]

FDMA, TDMA, CDMA and SDMA.

MODULE 5: Modulation schemes

[6 hours]

BPSK, QPSK and variants, QAM, MSK and GMSK, multicarrier modulation, OFDM.

MODULE 6: Multipath mitigation technique

[6 hours]

Diversity receivers- selection and MRC receivers, RAKE receiver,

Equalization: linear-ZFE and adaptive, DFE. Transmit diversity-Altamonte scheme.

MODULE 7: MIMO systems

[5 hours]

MIMO and space time signal processing, spatial multiplexing, diversity/multiplexing

Trade-off. Performance measures- Outage, average snr, average symbol/bit error rate.

MODULE 8: 3G and 4G Wireless Standards

[5 hours]

GSM, EDGE, GPRS, IS-95, CDMA 2000 and WCDMA.

Text/Reference Books:

1. WCY Lee, Mobile Cellular Telecommunications Systems, McGraw Hill, 1990.
2. WCY Lee, Mobile Communications Design Fundamentals, Prentice Hall, 1993.
3. Raymond Steele, Mobile Radio Communications, IEEE Press, New York, 1992.
4. AJ Viterbi, CDMA: Principles of Spread Spectrum Communications, Addison Wesley,
5. 1995.
6. VK Garg&JE Wilkes, Wireless & Personal Communication Systems, Prentice Hall,
7. 1996.

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1818PE31	Mixed Signal Design	3-0-0	3

Course Outcomes:

At the end of the course, students will be able to:

CO1: Identify the practical situations where mixed signal analysis is required.

CO2: Design Switched capacitor circuit and comparator circuits.

CO2: Analyze and handle the inter-conversions between signals.

CO3: Design systems involving mixed signals and its layout.

MODULE 1: Analog and discrete-time signal processing [10 hours]

Analog CMOS circuits, Basics of CMOS amplifiers: CS,CG,CD stages, Cascoded stages, Folded cascade Amplifier, MOS Current Mirror, MOSFET cascade current mirror, Differential Amplifiers. Introduction to sampling theory; Analog continuous time filters: passive and active filters; Basics of analog discrete-time filters and Z-transform.

MODULE 2: Switched-capacitor filters [6 hours]

Switched capacitor circuits, Switched capacitors amplifiers, Switched capacitor filters, Non idealities in switched-capacitor filters; Switched-capacitor filter architectures; Switched-capacitor filter applications.

MODULE 3: Comparators [4 hours]

Characterization of a comparator, open loop comparators, discrete time comparators, High speed comparators

MODULE 4: Basics of data converters [8 hours]

Successive approximation ADCs, Dual slope ADCs, Flash ADCs, Pipeline ADCs, Hybrid ADC structures, High-resolution ADCs, DACs.

MODULE 5: Mixed-signal layout [6 hours]

Interconnects and data transmission - Voltage-mode and current mode, signalling and data transmission; Current-mode signaling and data transmission.

MODULE 6: Introduction to frequency synthesizers and synchronization [6 hours]

Basics of PLL, Analog PLLs; Digital PLLs; DLLs.

Text/Reference Books:

1. R. Jacob Baker, CMOS mixed-signal circuit design, Wiley India, IEEE press, reprint 2008.
2. Behzad Razavi, Design of analog CMOS integrated circuits, McGraw-Hill, 2003.
3. R. Jacob Baker, CMOS circuit design, layout and simulation, Revised second edition, IEEE press, 2008.
4. Rudy V. dePlassche, CMOS Integrated ADCs and DACs, Springer, Indian edition, 2005.
5. Arthur B. Williams, Electronic Filter Design Handbook, McGraw-Hill, 1981.
6. R. Schauman, Design of analog filters by, Prentice-Hall 1990 (or newer additions).
7. M. Burns et al., An introduction to mixed-signal IC test and measurement by, Oxford university press, first Indian edition, 2008.

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1818PE32	Low Power VLSI	3-0-0	3

Course Outcomes:

At the end of this course students will be able to

CO1: Explain the need for low power in VLSI.

CO2: Estimate and analyse the power dissipation in VLSI circuits.

CO3: Explain the software design for low power in various level.

CO4: Design low voltage and low power CMOS circuits for various applications.

CO5: Derive the architecture of low power SRAM circuit.

MODULE 1: Physics of Power Dissipation in CMOS FET Devices [6 hours]

Physics of power dissipation in MOSFET devices, power dissipation in cmos, low power vlsi design: Limits

MODULE 2: Power Estimation [10 hours]

Modeling in signals, Signal Probability calculation, Probabilistic Techniques for signal activity estimation, Statistical Techniques, Estimation of Glitching power, Sensitivity Analysis. Power estimation using the input vector compaction, power dissipation in domino cmos, high level power estimation, Information theory based approaches, Estimation of maximum power.

MODULE 3: Synthesis for Low Power [6 hours]

Behavioral Level Transforms, Logic Level Optimization for Low power, Circuit Level Optimization.

MODULE 4: Design and Test of Low Voltage CMOS Circuits [10 hours]

Circuit Design style, Leakage current in deep submicrometer transistors, Deep submicrometer device design issues, Key to minimizing SCE, Low voltage circuit design techniques, Designing deep submicrometers with elevated intrinsic leakage, multiple supply voltages.

MODULE 5: Low Power Static RAM Architectures [8 hours]

Organization of a static RAM, MOS Static RAM Memory cell, Banked organization of SRAMs, Reducing voltage swings on bit lines, Reducing power in write driver circuits, Reducing power in sense amplifier circuits, method for achieving low core voltages from a single supply.

Text Books and References

1. K. Roy and S. C. Prasad, Low Power CMOS VLSI Circuit Design, John Wiley and Sons, 3rd Edition, 2009.
2. Jan Rabaey, Low Power Design Essentials, Springer Publications, 1st Edition, 2009.
3. Chandrakasan and R. Brodersen, Low-Power CMOS Design, IEEE Press, 1st Edition, 1995.
4. Chandrakasan, Bowhill, and Fox, Design of High-Performance Microprocessors, IEEE Press, 1st Edition, 2000.
5. G. Yeap, Practical Low Power Digital VLSI Design, Springer Publications, 1st Edition, 1995.

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1818PE33	Biomedical Electronics	3-0-0	3

Course Outcomes: At the end of this course students will be able to

CO1: Explain the principle of physiological systems and biomedical signals.

CO2: Apply electronic systems in biological and medical applications.

CO3: Develop the skills necessary to communicate findings and interpretations from medical imaging and recording systems.

CO4: Explain the use of therapeutic equipment.

MODULE 1: Physiological systems and Signals [6 hours]

Biology of the heart, circulatory and respiratory systems, auditory systems, physiology of nerve and muscle cells, fundamental organization of brain and spinal cord.

MODULE 2: Biosignals [6 hours]

Origin of bioelectric signals, electrocardiogram (ECG), phonocardiogram (PCG), encephalogram (EEG) and electromyogram (EMG). Spectral characteristic of biosignals.

MODULE 3: Physiological Transducers [10 hours]

Electrodes: silver-silver chloride electrodes, electrodes for ECG, EEG, EMG, Microelectrodes. Performance characteristics of transducers, classification of transducers based on Electrical principle involved: Resistive position transducer, resistive pressure transducer, inductive pressure transducer, capacitive pressure transducer; Self generating inductive transducer: linear variable differential transformer (LVDT), Piezoelectric Transducer. Transducers for body temp measurement, photoelectric transducers, pH measurement.

MODULE 4: Recording Systems [8 hours]

Preamplifier, Signal conditioning: Differential amplifier, current to voltage converter, instrumentation amplifier; biomedical filters: LPF, HPF, bandpass, band stop (Notch filter); 4-20ma transmitter, source of noise in low level measurement, Recording systems for ECG, PCG, EEG and EMG.

MODULE 5: Medical Imaging Systems [5 hours]

X-ray imaging, Computed tomography, ultrasonic imaging systems, Magnetic resonance imaging system, thermal imaging systems.

MODULE 6: Therapeutic equipment [5 hours]

Cardiac pacemaker, cardiac defibrillators, haemodialysis machine, lithotriptors, ventilators, bionic ear.

Text/ Reference Books:

1. L. Cromwell, F. J. Weibell, E.A. Pfeiffer. "Biomedical Instrumentation and Measurement" Pearson Education, 2003
2. R.S. Khandpur, "Handbook of Biomedical Instrumentation"
3. TATA McGRAW HILL, 2005
4. J. Enderle, S. Blanchard, J. Bronzino. "Introduction to Biomedical Engineering"
5. Academic Press, 2000
- 6.

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1818OE31	Machine Learning	3-0-0	3

Course Outcomes:

At the end of the course the students will be able to

CO1: Explain various concepts of machine learning hypothesis, evaluation and cross-validation methods.

CO2: Identify different types of learning techniques, algorithms, models.

CO3: Apply different models of machine learning on data.

CO4: Evaluate different models generated from data.

MODULE 1: [5 hours]

Basic definitions, types of learning, hypothesis space and inductive bias, evaluation, cross-validation.

MODULE 2: [8 hours]

Linear regression, Decision trees, over fitting, Instance based learning, Feature reduction, Collaborative filtering based recommendation, Bias Variance Trade-off, Generalization errors, model selection, evaluation metrics

MODULE 3: [8 hours]

Basic Probabilistic Modelling, Learning via Probabilistic Modelling, Probabilistic Models for Supervised Learning-Discriminative Approaches and Generative Approaches: Naïve Bayes, Logistic Regression, SupportVector Machine, Kernel function and Kernel SVM

MODULE 4: [7 hours]

Neural Network: Perceptron, multilayer network, gradient descent and back propagation, Convolutional Neural Networks, Recurrent Neural Networks, Deep Unsupervised Learning, Dynamic memory networks

MODULE 5: [6 hours]

Computational learning theory, PAC learning model, Linear Discriminant Analysis, Sample complexity, VC Dimension, Ensemble learning: Boosting and bagging, random forest, reinforcement learning

MODULE 6: [6 hours]

k-means clustering, Gaussian mixture model and EM algorithm

Textbooks/Reference Books:

1. Machine Learning. Tom Mitchell. First Edition, McGraw-Hill, 1997
2. Introduction to Machine Learning Edition 2, by Ethem Alpaydin

Course Code	Course Title	Hours per week L-T-P	Credit C
ECE1818OE32	Advanced Computer Architecture	3-0-0	3

Course Outcomes:

At the end of the course the students will be able to

CO1: classify instruction set architectures.

CO2: describe the operation of performance enhancements such as pipelines, dynamic scheduling, branch prediction, caches, and vector processors.

CO3: explain modern architectures such as RISC, Super Scalar, VLIW (very large instruction word), multi-core and multi-cpu systems.

CO4: develop applications for high performance computing systems.

MODULE 1:

[6 hours]

8085/8086 organization, I/O, Memory, Instruction set, Design of ALU.

MODULE 2:

[6 hours]

Instruction Set Principles and Examples, Classification of Instruction Set Architectures, Instruction Formats and Semantics, Memory Addressing Modes, Operations in the Instruction Set, Encoding and Instruction Set, the Role of Compilers.

MODULE 3:

[6 hours]

Advanced Pipelining and Instruction-Level Parallelism: Basic Pipeline Operations, Data and Control Pipeline Hazards, Instruction-Level Parallelism, Dynamic Instruction Scheduling and Branch Prediction.

MODULE 4:

[8 hours]

Memory-Hierarchy Design: Cache Design Issues, Performance Evaluation, Virtual Memory Addressing, Memory Protection Mechanisms, Memory coherency techniques.

MODULE 5:

[8 hours]

Storage Systems: Types of Storage Devices, Buses-Connecting I/O Devices to CPU/Memory, I/O Performance Measures, Reliability, Availability, and RAID, Interfacing to an Operating System.

MODULE 6:

[5 hours]

Thread Level Parallelism: Multiprocessor Systems and Applications, Centralized Shared-Memory Architectures, Distributed Shared-Memory Architectures, Execution Synchronization, Models of Memory Consistency.

MODULE 7:

[4 hours]

Data Level Parallelism: Vector Processing, Support for Multimedia Applications, Graphics Processing Units.

Books / References :

1. Computer Architecture: A Quantitative Approach, 5th Edition John L. Hennessy and David A. Patterson Morgan Kaufmann Publishers.
2. D. Sima, T. Fountain, P. Kacsuk, Advanced Computer Architecture, Addison-Wesley, 1997.
3. H.S. Stone, High-performance Computer Architecture, 3rd edition, Addison-Wesley, 1993.
4. Patterson, D. A. and Hennessy, J. L., Computer Organization and Design: The Hardware/Software Interface, Morgan Kaufmann, 1998.
